LLM Proxy-Assisted Combinatorial Auctions
A Modular Elicitation-Event Framework and Empirical Evaluation

Matthew Targett
MSc Computational Finance

August 3, 2026

Abstract

Combinatorial auctions allow bidders to express values for packages of goods, but their formal bid language and exponential bundle space can make participation cognitively and technically demanding. This paper develops a modular framework in which a person describes preferences in natural language, an LLM-based proxy constructs an initial combinatorial bid, and the auction mechanism emits elicitation events that identify economically salient bundles for exact refinement. The framework separates environment generation, preference interpretation, bid construction, mechanism feedback, and evaluation, and is instantiated in iterative sealed and ascending clock auctions. In 95 matched PC-component auction cases, interest-guided candidate generation reduces the number of initially considered bundle values by 55.5% relative to full powerset support. Mechanism-driven refinement raises mean allocation efficiency from 86.4% to 94.8% in the sealed format and 92.8% in the clock format, with the clock using fewer exact value queries. Payment reconstruction remains less accurate than allocation recovery. The results show how natural-language proxy assistance and mechanism feedback can be combined in a reproducible auction architecture, while also identifying the information that sparse elicitation leaves unresolved.

Keywords: combinatorial auctions; preference elicitation; large language models; proxy bidders; XOR bids; VCG; clock auctions; sealed auctions.

1 Introduction

Combinatorial auctions allocate heterogeneous goods when values depend on packages rather than isolated items. They can express complements, substitutes, and alternative configurations, but that expressiveness creates two barriers to participation. First, m goods induce 2^m possible bundles. Second, bidders must translate informal preferences into a formal bid language and reason about package values. Even where computation is available, this reporting task can impose substantial cognitive and technical burden.

LLM-based proxies offer a possible interface between a person and that formal market. A bidder can communicate in ordinary language, while a proxy interprets the disclosure, constructs a combinatorial bid, and responds to structured feedback [Huang et al., 2025]. Such assistance could make expressive auctions accessible to participants who would otherwise struggle to formulate XOR package bids. This paper does not conduct a human-subject study and therefore does not claim measured reductions in cognitive burden. Instead, it develops and tests the computational architecture required for that interface.

The central proposal is an *elicitation-event framework*. A proxy first converts natural-language evidence into a structured interest map and an initial set of provisionally valued bundles. During the auction, the mechanism emits events that expose allocation-relevant or payment-relevant uncertainty. The proxy maps each eligible event to a formal refinement action. This makes the mechanism an elicitation environment rather than merely a passive consumer of a one-shot LLM bid. The organising information path is

$$\underbrace{\text{support discovery}}_{\text{what bundles are considered}} \longrightarrow \underbrace{\text{value estimation}}_{\text{how they are initially ranked}} \longrightarrow \underbrace{\text{mechanism-driven refinement}}_{\text{which estimates are corrected}}.$$

Two contrasting mechanism families are evaluated:

1. an **iterative sealed XOR auction** with provisional feedback and final VCG payments;
2. an **ascending clock mechanism** with supplementary XOR atoms and final VCG payments.

They share the same initial proxy state and formal bid representation but expose different evidence. The sealed mechanism repeatedly reveals provisional allocations and counterfactual competitors. The clock produces a price and demand trajectory that restricts the terminal verification frontier. Studying both formats tests whether the event abstraction can accommodate distinct mechanism-specific routes to economically salient bundles.

The event interface is designed to be extensible to other mechanisms, but the empirical evidence is narrower: it covers two implemented mechanism families, one generated PC-component master population, and one frozen model configuration. The paper therefore distinguishes architectural generality from empirical generalisation throughout.

The main research question is:

> How can an LLM proxy translate natural-language preferences into a tractable combinatorial bid, and how effectively can mechanism-specific elicitation events refine that bid toward efficient allocation?

The primary contribution is the modular elicitation-event interface. Three supporting contributions make that interface testable and operational:

1. **A modular elicitation-event interface.** Sealed and clock mechanisms expose different economic signals through a common event abstraction, allowing targeted, deduplicated refinements and paired evaluation from the same frozen initial state.
2. **Interest-guided candidate support and provisional initialisation.** A person-specific interest map determines what the proxy considers; provisional values then rank that support. This separates omission from scoring error.
3. **Structured test environments.** A constrained pipeline produces heterogeneous private preferences and deterministic full-information benchmarks for auditable evaluation.
4. **Information-aware diagnostics.** Candidate coverage, allocation trajectories, event incidence, and VCG witness logging identify whether information is lost during support selection, provisional scoring, allocation refinement, or payment-counterfactual construction.

Caching, chunked structured output, fail-closed parsing, frozen replay, and detailed provenance are supporting experimental controls rather than independent conceptual contributions.

2 Background and Literature Review

This project lies at the intersection of classical preference elicitation, iterative and clock combinatorial auctions, machine-learning-powered elicitation, and recent work on LLM-based proxies.

2.1 Classical Preference Elicitation

Combinatorial auctions are appropriate when bidders have complements or substitutes across goods. Their central obstacle is communication. Nisan and Segal show that, for general valuation classes, efficient allocation can require exponential communication [Nisan and Segal, 2006]. This result motivates restricted bid languages, structured valuation classes, approximation, and selective elicitation.

Lahaie and Parkes connect preference elicitation to computational learning theory. They show how value queries can simulate membership queries and how certain demand-style queries can play the role of equivalence-query counterexamples [Lahaie and Parkes, 2004]. This gives formal query-complexity guarantees for some structured valuation classes and bid languages, but it still assumes that bidders can answer formal economic queries. In practice, value and demand queries may be cognitively demanding.

Work on structured classes such as strong substitutes illustrates the same tension. Goldberg, Lock, and Marmolejo-Cossío study learning strong-substitutes demand via queries and show efficient algorithms for restricted settings but hard lower bounds in more general cases [Goldberg et al., 2022]. The theoretical lesson is that structure matters. The practical question is how to extract useful structure from human preference descriptions without requiring full formal revelation.

2.2 Iterative and Clock Auction Design

Iterative combinatorial auctions reduce upfront communication by eliciting information over rounds. Parkes surveys mechanisms that alternate between provisional allocations, price updates, and new bids [Parkes, 2006]. iBundle is an early ascending package auction in which prices and provisional allocations guide bidders towards welfare-relevant packages [Parkes and Ungar, 2000]. Such mechanisms motivate the iterative sealed arm in this project: each round solves a provisional WDP and uses the result to select bundles for refinement.

Clock auctions provide a different information structure. They post item prices, ask bidders for demanded bundles, and raise prices on over-demanded goods. In a clock setting, the proxy observes a price path and a sequence of revealed demands rather than only provisional winners and losers. These revealed bundles can subsequently delimit a sparse supplementary and counterfactual verification frontier.

The clock mechanism in this paper is CCA-inspired but not a full production CCA. It uses ascending item prices, top-surplus demand, accumulated supplementary XOR atoms, and final VCG payments. It does not implement activity-constrained supplementary bidding, optimised price increments, or core-selecting payments. This is intentional: the clock is used as a price-discovery environment whose revealed-demand path restricts later exact verification.

2.3 Machine-Learning-Powered Elicitation

Machine-learning-powered combinatorial auctions use partial bidder responses to train surrogate valuation models and select future queries. MLCA uses machine learning to guide value-query selection and reports high efficiency with comparatively few elicited values [Brero et al., 2019]. iMLCA extends this idea by allowing interval bids and activity rules to reduce the reporting burden associated with exact values [Lubin et al., 2020].

More recent ML-powered clock mechanisms emphasise demand queries. Soumalias et al. propose an ML-powered combinatorial clock auction that learns from demand-query information and selects queries with high clearing potential [Soumalias et al., 2023]. Later work combines prices, bids, and values in a hybrid auction design [Soumalias et al., 2024]. This literature establishes welfare-versus-query-cost evaluation as a natural empirical standard and shows that query type matters. The present project differs by replacing direct human interaction with a proxy-mediated natural-language interface.

2.4 LLM-Based Proxies

Huang et al. propose LLM-based proxies for accelerated preference elicitation in combinatorial auctions [Huang et al., 2025]. Their proxies interact with a mechanism on behalf of a bidder, maintain internal state, and use natural-language questions, value queries, demand queries, and learned bid representations. This is the closest conceptual predecessor to the present work. It motivates the distinction between a person model and a proxy model, and it demonstrates that LLMs can reduce the burden of translating qualitative preferences into formal bids.

Soumalias et al. study LLM-powered preference elicitation in combinatorial assignment [Soumalias et al., 2025]. Their work similarly highlights the importance of protocol design, query cost, and LLM response variability.

The gap addressed here is mechanism portability. Existing LLM-proxy work is largely mechanism-specific. This paper asks how one event-driven proxy architecture behaves when embedded in sealed and clock mechanisms that expose different feedback signals.

3 Formal Framework

3.1 Combinatorial Auctions and XOR Bids

Let $G = \{g_1, \ldots, g_m\}$ be a finite set of indivisible goods, and let $\mathcal{B} = 2^G$ be the bundle space. There are bidders $N = \{1, \ldots, n\}$. Bidder i has a hidden valuation $v_i : \mathcal{B} \to \mathbb{R}_{\geq 0}$. An allocation is $a = (a_1, \ldots, a_n)$, where $a_i \subseteq G$, subject to feasibility:

$$a_i \cap a_j = \emptyset \quad \forall i \neq j.$$

The full-information efficient allocation is

$$a^* \in \arg\max_a \sum_{i \in N} v_i(a_i).$$

The implementation represents formal bids using XOR atoms. An XOR bid for bidder i is a finite set

$$\theta_i = \{(b, \hat{v}_i(b)) : b \in B_i\},$$

where $B_i \subseteq 2^G$. The induced reported valuation is

$$v^{\theta_i}(b) = \max_{\substack{b' \in B_i \\ b' \subseteq b}} \hat{v}_i(b'),$$

with value zero if no reported atom is contained in b. This free-disposal interpretation is used by the WDP solver.

Given reported bids $\theta = (\theta_1, \ldots, \theta_n)$, the WDP is

$$\max_a \sum_i v^{\theta_i}(a_i)$$

subject to feasibility. The implementation solves this problem by an integer linear programming formulation over XOR atoms.

3.2 Auction Environment and Person Model

An auction environment is

$$\mathcal{E} = (G, N, D_A, D_G, (\Lambda_i)_{i\in N}, (v_i)_{i\in N}),$$

where D_A is the auction description, D_G maps goods to public text descriptions, Λ_i is bidder i's private structured preference profile, and v_i is the deterministic hidden benchmark valuation induced by that profile. The bidder proxy observes the public catalogue and the person's disclosed answer, but never observes Λ_i or v_i. The person simulator and evaluator may access the private profile for different purposes.

The person simulator is distinct from the bidder proxy. Before an experiment, the profile is rendered deterministically into a brief qualitative disclosure contract D_i. A person model conditioned on D_i answers one canonical opening question, disclosing natural-language goals, positively valued options, relevant alternatives or complements, acquisition mode, explicit exclusions, and an approximate total budget. The answer is independently verified against Λ_i before it is accepted.

During the principal mechanism runs, a value query for bundle b is answered deterministically as $v_i(b)$. This removes repeated LLM response noise and isolates the bundle-selection policy induced by the auction mechanism. LLM-based value and demand answers remain available as robustness treatments, but they are not used in the main scalability specification.

The implementation records three model roles separately: environment, person, and bidder proxy. The environment and person roles may use the same model, while the bidder proxy can use a different provider and model. In the current frozen population the environment and person roles use `gpt-5.6-sol`, while interest-map and provisional-valuation inference use `gemini-3.6-flash`. Role separation is an information boundary, not necessarily a requirement that all roles use different model families.

3.3 Structured Environment Generation

The environment pipeline begins from a human-specified population design, not an unconstrained request for an auction. The design fixes the 16 good identifiers and descriptions, their functional categories, 16 bidder archetypes across four broad strata, and population- and sample-level constraints. It deliberately leaves bidder-specific numeric and structural preferences to the environment model.

Each bidder is generated from a latent profile

$$\Lambda_i = (w_i, \mathcal{S}_i, \mathcal{C}_i, B_i, \rho_i, D_i),$$

where w_i gives base item values, $\mathcal{S}_i$ person-specific substitute groups, $\mathcal{C}_i$ complement groups, B_i a budget cap and scale, ρ_i saturation behaviour, and D_i a role and identity description. Every good is also classified as core, secondary, low interest, or zero value for that bidder.

For a substitute group $S \in \mathcal{S}_i$, the profile gives full value to the highest-valued selected item and discounted backup or resale value to additional selected items:

$$\mathrm{sub}_i(S, b) = w_i(g^*) + \lambda_{iS} \sum_{g\in(b\cap S)\setminus\{g^*\}} w_i(g), \quad g^* \in \arg\max_{g\in b\cap S} w_i(g).$$

A complement group $C \in \mathcal{C}_i$ contributes a bonus α_{iC} when $C \subseteq b$. For ordinary single-system buyers, $\lambda_{iS} = 0$ implements a `choose_one` relationship. For resellers, integrators, or multi-machine buyers, a positive backup factor implements `can_use_multiple`. Thus functional similarity does not imply that joint ownership is valueless. The generated value is a sum of ordinary item contributions, substitute-group contributions, complement bonuses, and saturation or cap adjustments. Values are repaired to satisfy free disposal.

The environment model generates the structured profile rather than directly enumerating a value for every bundle. Deterministic code then stores complete hidden values for every non-empty bundle. For m goods this gives $2^m - 1$ benchmark values per bidder. These tables support full-information allocation, exact value queries, welfare evaluation, and diagnostic decomposition.

Generation proceeds in batches with bounded automatic repair. A batch is accepted only if its profiles satisfy schema, classification, monetary, and person-level interest-density constraints. The assembled population is then validated for per-good positive and core bidder coverage, cross-stratum competition, multiway substitute structure, complement prevalence, and acquisition-mode consistency.

3.4 Coverage-Stratified Environment Sampling

Experiments use a fixed validated 16-by-16 master population but vary the selected goods and bidders. For each seed, a deterministic coverage-stratified search creates one nested ordering of all goods and one nested ordering of all bidders. Prefixes of these latent orderings generate the required sizes while the public scenario preserves catalogue declaration order.

Every scalability composition is validated for minimum interest coverage, positive value for each selected bidder, complement and substitute prevalence, category and bidder-stratum diversity, at least two full-information winners, and a bound on the largest winner's welfare share. The five selected composition seeds produce distinct good sets, bidder sets, and combined environments for every experimental cell. They are alternative nested orderings of one master population, not independently generated populations.

3.5 Event-Driven Proxy State

Bidder i's proxy has state

$$S_i = (K_i, I_i, \mathcal{C}_i, O_i, \tilde{v}_i, \theta_i),$$

where K_i is the knowledge base, I_i the interest map, $\mathcal{C}_i$ the candidate bundle support, O_i explicit observations, $\tilde{v}_i$ provisional or refined value estimates, and θ_i the mechanism-facing XOR bid.

An elicitation event is represented as

$$e = (\tau, i, b, x, \rho),$$

where τ is the event type, i the target bidder, b an optional target bundle, x the mechanism state that triggered the event, and ρ its eligibility evidence or economic rationale. This representation separates three operations:

$$E_t = \mathrm{Emit}_M(x_t), \qquad q_t = \mathrm{Select}_\pi(E_t, S_t), \qquad S_{t+1} = \mathrm{Update}(S_t, q_t, y_t),$$

where the mechanism emits eligible events, the event policy selects an action, and the proxy updates its state after response y_t. A policy may prioritise or ignore emitted events without changing the auction implementation, while a new mechanism can emit the same event type from different state evidence.

The proxy responds to elicitation events:

$$\pi_i : \mathcal{E}_{event} \times S_i \to (\mathcal{A}_{proxy}, S'_i).$$

As a general interface, events may trigger language interaction, structural inference, value or demand refinement, bid submission, or no action. This expressiveness should not be confused with the experimental instantiation in this paper. Here the opening exchange, interest map, candidate support, and provisional values are frozen before the auction. At auction time, a deterministic event policy selects an unqueried bidder–bundle pair, exact value lookup deterministically returns the hidden benchmark value, and that value replaces the provisional report. The runner does not deliberate over a transcript, autonomously choose a new query type, or make any auction-stage LLM call. The proxy state therefore defines a reusable general architecture, while the experiments evaluate a deliberately non-agentic event-policy instantiation.

The generic update is

$$e_t \to \pi_i(e_t, S_{i,t}) \to q^V_{i,t}(b) \to v_i(b) \to S_{i,t+1} \to m_{i,t+1}.$$

For sealed auctions, $m_{i,t+1}$ is an XOR bid. For clock auctions, it is a demand response and supplementary XOR information.

4 Proxy Inference Pipeline

4.1 Natural-Language Questioning

The current main configuration uses one canonical opening question tailored to the PC-component catalogue. It asks what the person hopes to obtain, how the alternatives fit their needs, whether any goods are useful only together or remain useful in multiples, and approximately how much they would spend overall. The same question is used across bidders in a given catalogue so that differences in their answers arise from preferences rather than question wording.

The person answer is intentionally brief and qualitative. It is not intended to recover a complete value table. It discloses enough information to support candidate selection and approximate scale without exposing base values, complement bonuses, or the hidden bundle table. The answer and its preparation-time verification record are frozen before either auction mechanism is run. Proxy-generated opening questions remain an optional robustness treatment.

4.2 Interest Map

The proxy derives a lightweight interest map from the auction description, item descriptions, natural-language question, and person answer:

$$I_i = (G_i^+, G_i^-, C_i, S_i, \beta_i, r_i).$$

Here G_i^+ is the inferred interested item set, G_i^- excluded items, C_i complement groups, S_i substitute groups, β_i a budget hint, and r_i a rationale.

Each inferred substitute group additionally records an acquisition mode

$$\mu(S) \in \{\texttt{choose\_one}, \texttt{can\_use\_multiple}, \texttt{unclear}\}$$

and textual evidence. A restrictive `choose_one` mode is accepted only when the answer states that at most one member is useful or that additional members add no meaningful value. Ranking one good above another, calling an item a fallback, or observing functional similarity is insufficient. This distinction prevents a GPU category, for example, from being incorrectly treated as exclusive for a reseller who can use multiple cards.

The main implementation uses a closed-world map: every catalogue item must be classified, and an item that is neither positively disclosed nor included in a disclosed structural group is placed in G_i^-. Candidate generation considers subsets of interested items, never includes excluded items, and filters joint bundles only when they contain multiple members of an evidence-grounded `choose_one` group. `Can_use_multiple` and `unclear` groups are retained conservatively.

The implementation enforces only generic consistency. It does not impose hidden PC-build-specific rules or use the private profile to repair the proxy's map. Hidden truth is used solely for diagnostics: item precision/recall, exact group precision/recall, acquisition-mode accuracy, choose-one precision/recall, dangerous false exclusivity, and oracle-versus-inferred candidate-set precision and recall.

The current implementation also avoids silent full-powerset fallback in live experiments. If interest-map parsing fails after retries, the run can fail closed under an interest-map failure policy. A degraded all-items fallback remains available for debugging, but it is not the preferred experimental setting because it can generate excessively broad and incoherent support.

4.3 Candidate Bundle Generation

Given I_i, candidate generation produces a priority-ordered set

$$\mathcal{C}_i = \Psi(I_i, G, M_{\text{cand}}).$$

The implemented ordering is:

1. complete declared complement groups;
2. singletons of interested items;
3. all remaining valid subsets in ascending size and lexicographic order.

A bundle is valid if it does not include multiple goods from the same declared `choose_one` substitute group. Bundles containing multiple members of `can_use_multiple` or `unclear` groups remain eligible. A candidate cap M_{cand} can be applied after prioritisation, although the principal scalability preparation uses $M_{\text{cand}} = \infty$ and therefore leaves the inferred support uncapped.

This design intentionally keeps the interest map simple. Its main function is to reduce support size while preserving important complement groups and singletons. Failure modes are logged through candidate-count diagnostics, candidate reduction ratios, coverage measures, and interest-map quality flags such as broad interest sets with no substitute groups.

4.4 Provisional Valuation and Chunking

After candidate generation, the proxy assigns provisional values:

$$\tilde{v}_i(b), \quad b \in \mathcal{C}_i.$$

The proxy model receives the public catalogue, opening exchange, interest map, and enumerated candidate support. It returns one estimated value for each candidate bundle. These values both rank the support and initialise the proxy's mechanism-facing XOR bid. They are proxy-side inferences, not person value queries, and they do not have access to the private profile or hidden valuation table.

For large supports, the implementation uses deterministic PV chunking. If

$$|\mathcal{C}_i| > K_{\text{PV}},$$

candidate bundles are split into ordered chunks of at most K_{PV} bundles. Each chunk is valued by a separate provisional-valuation call and the results are merged. Chunking preserves the semantics of the initial bulk PV stage while reducing the risk of truncation or malformed large JSON outputs. PV chunks are counted as shared initial proxy inference calls, not value-query refinements.

The current preparation configuration uses chunks of at most 64 bundles and fail-closed PV behaviour. If a chunk fails after parse retries, the run aborts with a diagnostic error identifying the bidder, candidate count, chunk size, token budget, and original parsing exception. A degraded zero-initialisation policy exists for debugging, but failed PV never triggers mass direct value queries over the candidate support. Otherwise a structured output failure would silently change the elicitation treatment and contaminate person-query counts.

4.5 Provisional-Value Calibration

Provisional valuations are estimated from the person's short qualitative disclosure alone, and the estimator's error need not be centred. The implementation therefore supports an explicit calibration of the raw provisional value, parameterised by a family, a multiplicative scale, and an optional bundle-size adjustment:

$$\hat{v}^{cal}(b) = \min\Big\{\bar{B},\ s \cdot \hat{v}^{raw}(b) \cdot \gamma^{\max(0,|b|-k_0)}\Big\}.$$

Here $s > 0$ is the scale, $\gamma > 0$ the size factor, k_0 the size threshold, and $\bar{B}$ the bidder's *disclosed* budget, taken from the elicited interest map rather than from any hidden value. The scale is not restricted to $(0, 1]$: the estimator's qualitative conservatism instructions mean the observed bias is typically downward, so the corrective scale is generally greater than one. The family selects which terms apply: the identity (raw values), scale only, or scale with the size adjustment.

Calibration applies only to initial LLM-inferred provisional values. Exact deterministic value-query answers and mechanism-triggered refinements are ground truth and are reported unchanged.

Frozen elicitation packs store raw, uncalibrated provisional values, so alternative calibration rules are replayed over an identical elicitation without new LLM calls; a calibrated and an uncalibrated arm therefore differ only in the calibration and not in re-elicitation noise. Every run records its fully resolved effective calibration, together with a stable configuration hash, in its console header, machine-readable run configuration, and result tables.

The final specification uses the accepted uniform scale $s = 1.4758$, a disclosed-budget cap, and no bundle-size adjustment ($\gamma = 1$). It was selected before the main mechanism runs using nine synthetic consumer-bundle environments spanning camera/video, kitchen, and travel domains, with three independently generated instances per domain. Three folds leave out one instance index across all domains, fitting on six environments and evaluating on the remaining three. The objective is mean absolute reported-VCG payment error normalised by optimum welfare, subject to a mean-efficiency tolerance. This procedure shares no goods or bidder profiles with the PC-component population. On the nine held-out evaluations, calibration reduces the objective from 0.289 to 0.233 and improves seven of nine cases. The accepted calibration and its hash are frozen in the final experiment specification.

Algorithm 1 Natural-language disclosure to initial XOR bid

Require: Public catalogue G, canonical question, person disclosure contract D_i
Obtain and independently verify opening answer A_i
Infer interest map I_i from (G, A_i)
Generate $\mathcal{C}_i = \Psi(I_i, G, M_{\text{cand}})$
Estimate $\tilde{v}_i(b)$ for every $b \in \mathcal{C}_i$, using deterministic chunks if required
Apply the frozen calibration rule and disclosed-budget cap
return $\theta_i^0 = \{(b, \tilde{v}_i(b)) : b \in \mathcal{C}_i\}$ and frozen proxy state S_i^0

4.6 Caching, Freezing, and Replay

LLM calls are cached by a hash of the rendered prompt and all response-relevant metadata. Cache modes include off, read-write, read-only, and refresh. Each successful response is committed independently, allowing interrupted preparation to resume without repeating completed calls.

Caching supports preparation, but the principal experimental control is the frozen elicitation pack. A pack contains the accepted opening exchange, interest map, candidate bundles, raw provisional values, model provenance, generation settings, and call records for one catalogue and bidder set. Fresh mutable proxies replay this immutable initial state with zero LLM calls.

Consequently sealed and clock arms differ only in their mechanism paths and deterministic refinement queries.

Initial elicitation depends on the selected goods but not on the number of rival bidders. For each seed, preparation therefore generates one master pack for every goods count and then projects ordered bidder subsets into the 19 scalability cells. This avoids regenerating the same bidder-catalogue exchange for the fixed-goods bidder sweep.

5 Mechanism Environments

5.1 Iterative Sealed XOR Auction

At round r, each proxy submits an XOR bid θ_i^r, and the mechanism solves:

$$a^r \in \arg\max_a \sum_i v^{\theta_i^r}(a_i).$$

The WDP and reported allocation provide the mechanism state from which events are emitted; the auction algorithm itself does not prescribe which bundle is queried. After each set of event responses, corrected XOR atoms replace their provisional counterparts and the WDP is solved again. The main termination condition is the absence of any eligible new refinement after at least one cycle. Forty rounds is a non-binding runaway safeguard in the final suite. Every bidder–bundle query is deduplicated globally.

5.2 Sealed Elicitation Events

Four event families are considered because they expose different parts of the reported allocation frontier.

Incumbent verification. The provisionally allocated bundle is the atom whose error can most directly invalidate the reported winner. Whenever a bidder receives a non-empty, previously unqueried bundle in the current WDP allocation, the event targets that bidder–bundle pair for exact value lookup.

Winner-removal allocation counterfactual. VCG payments and competitive displacement depend on allocations that become feasible when a current winner is absent. For each winning atom, the mechanism temporarily removes that atom, re-solves the WDP, and emits events for previously unqueried bundles that enter the counterfactual allocation.

Scarcity-avoiding fallback. A high provisional incumbent may occupy contested goods even when its bidder has a credible alternative. If an incumbent contains currently contested goods, the event selects the bidder's highest-reported unqueried alternative that reduces contested-good exposure and retains at least 60% of the incumbent's reported value. It is triggered at most once per bidder.

Large-correction follow-up. A large initial error is evidence that a nearby provisional ranking may also be unreliable. If exact lookup changes a value by at least 25% under the symmetric relative correction measure, the event targets one unqueried structurally neighbouring candidate bundle. It is triggered at most once per bidder.

Independent loser exploration is disabled in the principal experiments: losers are queried only when they enter a winner-removal allocation. Section 8.4 selects among combinations of the four events; the definitions above do not presuppose that every event is empirically useful.

Algorithm 2 Iterative sealed elicitation

Require: Frozen initial bids θ^0, sealed event policy π_S

for round $r = 0, 1, \ldots$ **do**
  Solve the reported WDP to obtain a^r
  $E_r \leftarrow \text{Emit}_S(a^r, \theta^r)$
  Remove previously queried bidder–bundle pairs from E_r
  $Q_r \leftarrow \text{Select}_{\pi_S}(E_r, S^r)$
  if $Q_r = \emptyset$ after at least one elicitation cycle **then**
    break
  end if
  for all $(i, b) \in Q_r$ **do**
    Replace $\tilde{v}_i(b)$ by exact $v_i(b)$
  end for
end for
return final reported allocation and reported-space VCG payments

5.3 Ascending Clock with Supplementary XOR Bids

The clock arm posts item prices p_t. The reported surplus of bundle b for bidder i is

$$u^{\theta}_{i,t}(b) = v^{\theta_{i,t}}(b) - \sum_{g \in b} p_{t,g}.$$

Each proxy reports up to k top-surplus bundles; the principal specification uses $k = 3$. The mechanism increments every good with positive excess demand by \$50. All top-k demands observed along the price path are accumulated as supplementary XOR atoms. No exact value query is made while prices ascend: price discovery is therefore allowed to complete on the frozen calibrated reports. The clock stops when there is no excess demand; 500 rounds is a runaway safeguard rather than a target.

Thus price discovery is query-free and naturally terminates before terminal exact refinement begins. Let R_i denote bidder i's bundles revealed among its top-k demands. These revealed sets delimit the clock-specific event space: unrevealed losing bundles do not become eligible merely because they have high provisional values.

5.4 Clock Elicitation Events

Single-pass revealed VCG witness. Before any correction, the mechanism removes each provisional winner and solves the associated supplementary counterfactual allocation. This freezes a pre-query witness frontier. An event is emitted only when a counterfactual bundle is both unqueried and belongs to its bidder's revealed-demand set R_i. The event targets payment-relevant competitors while preserving the support discipline supplied by clock price discovery.

Winner closure. Corrections can change the supplementary allocation. After each re-solve, an event is emitted for every newly allocated, previously unqueried bundle. The mechanism alternates exact lookup and re-optimisation until all current winners have been verified. This targets allocation correctness rather than broad frontier exploration.

Staged revealed-witness closure. Winner corrections can expose new bidder-removal competitors. The mechanism therefore recomputes the revealed witness frontier after winner closure, queries eligible revealed witnesses, and returns to winner closure. The two stages alternate until neither produces a new query. Section 8.4 evaluates this sequence against narrower and broader terminal policies.

Algorithm 3 Clock price discovery and terminal revealed-winner verification

Require: Frozen bids θ^0, top-k rule, price increment Δp
while some good has positive excess demand **do**
    Record each bidder's top-k positive-surplus demands in R_i
    Raise each over-demanded good's price by Δp
end while
Construct supplementary XOR bids from all revealed demands
Freeze and query eligible pre-correction revealed VCG witnesses
repeat
    Query newly allocated unverified bundles until winner closure
    Recompute and query eligible revealed bidder-removal witnesses
until neither stage emits an unqueried bidder–bundle pair
return final supplementary allocation and reported-space VCG payments

5.5 Payments and the Information Required by VCG

VCG is retained as a common reported-space payment rule so that sealed and clock allocation paths remain comparable. Payment accuracy is analytically distinct from allocation efficiency. Even if every finally allocated bundle has been queried exactly and the efficient allocation has been found, bidder i's payment depends on the welfare attainable by the other bidders when i is removed. Sparse provisional reports can therefore yield accurate allocations but inaccurate winner-removal counterfactual welfare and revenue.

Allocation efficiency is consequently the primary mechanism outcome. Reported VCG revenue, full-information VCG revenue, and per-winner payment error are reported as secondary diagnostics. A post-allocation payment-counterfactual audit is a natural extension, but replacing VCG with pay-as-bid would conflate preference elicitation with strategic bid shading and is not part of the principal design.

Table 1: Feedback signals by mechanism.

Mechanism	Main feedback	Proxy refinement opportunity
Sealed XOR–VCG	Provisional WDP, allocated bundles, winner-removal frontier, contested goods	Refine current winners, counterfactuals, scarcity fallbacks, and one-step correction neighbours
Clock with supplementary VCG	Ascending prices, top-three revealed demands, terminal winner-removal frontier	After price discovery, close newly winning bundles and revealed VCG witnesses

6 Implementation

The repository separates deterministic auction logic from preparation-time LLM orchestration. Pure auction modules receive typed proxy states and emit typed events; they do not call model providers. Preparation modules generate and validate environments, disclosures, interest maps, and provisional values, then write immutable elicitation packs. Experiment runners replay those packs, solve XOR WDPs, answer selected exact queries, and record trajectories and VCG witnesses. This separation permits paired mechanism experiments without repeated model sampling.

Every preparation call and auction refinement is auditable through structured JSON, CSV, JSONL, and SQLite artifacts. The main text reports only controls needed to interpret the experiment; model identifiers, token limits, parse and retry behaviour, cache semantics, frozen policy parameters, artifact paths, and reconstruction commands are collected in Appendix A.

7 Experimental Design

7.1 Research Questions

The empirical design addresses:

1. Does the structured generation pipeline produce diverse, competitive, and economically non-trivial environments across the scalability grid?
2. How much does interest-map inference reduce candidate support relative to full powerset reporting?
3. How much welfare do calibrated provisional bids recover before exact refinement?
4. Which sealed and clock event policies improve allocation efficiency, and at what exact-query cost?
5. How do support, allocation efficiency, and exact-query burden scale with goods and bidders?
6. Why can reported-space VCG payments remain inaccurate when the allocation is efficient, and which event types reach payment witnesses?

7.2 Environments

The master population contains 16 PC-component goods and 16 bidders in four strata: gaming/performance, budget/office, professional/AI, and reseller/procurement. The bidder set includes enthusiast and value gamers, a streamer, an overclocker, student and office users, a family buyer, a minimalist user, an AI researcher, a creative professional, a software developer, a data analyst, a component reseller, a small-business IT buyer, a home-server operator, and a system integrator.

For each of five seeds, experiments follow three paths:

1. hold bidders at eight and vary goods from four to ten;
2. hold goods at eight and vary bidders from four to ten;
3. vary goods and bidders jointly from 4×4 to 10×10.

The common 8×8 anchor is counted once, giving 19 environments per seed and 95 environments overall. Within a seed, successive selections are nested. Across seeds, every cell has five distinct good sets, bidder sets, and combined compositions. These five composition seeds measure sensitivity to sampling from the fixed master population; they do not estimate variation across independently generated populations or auction domains.

7.3 Experimental Comparisons

The full-information allocation is an oracle benchmark rather than a feasible communication treatment. The final analysis distinguishes:

1. the analytical full-support burden $\sum_i(2^{|G|} - 1)$;
2. the final inferred support after item exclusion and evidence-grounded `choose_one` filtering;
3. the static allocation induced by calibrated provisional XOR bids on that support;
4. the final sealed and clock allocations after deterministic exact refinement.

The mechanism comparison starts sealed and clock arms from the same frozen interest-map-plus-PV state. Policy selection is reported separately through five-seed 8×8 ablations; the selected policies are then frozen for the 95-case scalability evaluation.

7.4 Metrics

The main outcome is true welfare:

$$W^{true}(a) = \sum_i v_i(a_i).$$

The global full-information optimum is

$$W^* = \max_a \sum_i v_i(a_i),$$

and global efficiency is

$$\eta(a) = \frac{W^{true}(a)}{W^*}.$$

The mechanism-specific full-information outcome a_M^{FI} is also computed where appropriate, giving

$$\eta_M^{rel}(a) = \frac{W^{true}(a)}{W^{true}(a_M^{FI})}.$$

The implementation records reported welfare, true welfare, reported-space VCG revenue, full-information VCG revenue, and true surplus. Communication diagnostics include exact value queries, preparation LLM calls and tokens, PV chunks, candidate bundles, and supplementary atoms. Mechanism diagnostics include clock rounds, event usefulness, winning-bundle coverage, final-allocation hits, termination reasons, and failure classifications. Payment analysis separates allocation error from counterfactual payment error.

The reported costs must not be conflated. Table 2 separates bidder interaction from offline machine preparation. Exact-query counts measure formal preference requests selected by the mechanism, but the principal experiments answer them by deterministic lookup. They are therefore a protocol-burden measure, not direct evidence of human effort.

7.5 Candidate and Interest-Map Diagnostics

Because full hidden valuation tables are available, candidate-support quality can be measured directly. Candidate reduction is

$$1 - \frac{|\mathcal{C}_i|}{2^{|G|} - 1}.$$

Winning-bundle coverage checks whether full-information allocation bundles are present in candidate supports. Top-k coverage compares each candidate set with the bidder's top hidden-value bundles. Interest-map diagnostics compare inferred and oracle item sets, substitute groups, acquisition modes, and resulting candidate supports.

Table 2: Information and cost layers in the experimental architecture.

Layer	Reported measure	Interpretation
Person interaction	One opening answer; number of exact VQs	Formal requests a deployed bidder would face
Initial formal support	Candidate bundle count	Bundle values the proxy must initially score
Machine preparation	LLM calls, tokens, chunks, latency	One-off frozen-pack construction cost
Auction computation	WDP solves and clock rounds	Deterministic mechanism runtime
Human cognitive burden	Not directly measured	Requires a human-subject study

The static initial-PV allocation and its full-information welfare measure the combined consequence of support selection and provisional scoring. Detailed diagnostics still distinguish omitted-bundle failure from miscalibration: a proxy may fail because it did not consider a relevant bundle, because an incorrect acquisition mode filtered it, or because it considered the bundle but assigned an inaccurate provisional value.

7.6 Frozen Specification and Reporting Protocol

The complete population and all 95 planned compositions pass deterministic population, structural, and economic preflight validation. Frozen elicitation packs were generated once per seed and goods count, then reused across every mechanism treatment. The final specification fixes the calibration scale at 1.4758; the sealed policy at competitive feedback with incumbent verification, winner-removal counterfactuals, scarcity fallbacks, and large-correction follow-up; and the clock at top-three demand, a $50 price step, natural no-excess-demand clearance, and terminal revealed-witness/winner closure. Model-role robustness is deliberately deferred rather than mixed into the primary dataset.

Plots and tables report the mean and dispersion across five seeds for each cell rather than joining one point from each nested path as if it were an independent trend. Because the 19 cells within a seed are nested and correlated, uncertainty intervals are computed from five seed-level means using a Student-t critical value; the 95 cells are not treated as 95 independent replications. The principal tables include:

1. environment, calibration, and frozen-pack validation;
2. full-powerset and final inferred candidate support;
3. static, sealed, and clock welfare, efficiency, queries, and rounds;
4. policy ablations and event-type frequency, correction magnitude, and VCG-witness incidence;
5. reported versus full-information VCG payment diagnostics.

8 Results

The final analysis contains 95 matched auction cases: 19 scalability cells for each of five composition seeds. Every sealed–clock pair starts from the same frozen disclosure, inferred support, raw provisional values, and calibration. Unless stated otherwise, reported percentages are equally weighted case means. Confidence intervals use the five seed-level means as the independent observations.

8.1 Generated Environment Structure

The accepted master population contains 16 goods in eight functional categories and 16 bidder archetypes in four strata. Figure 1 summarises its competitive and economic structure. A bidder assigns positive base value to 78.1% of the master catalogue on average, with bidder-level density ranging from 43.8% to 100%. Goods attract between nine and 16 positive bidders (mean 12.5), so exclusions coexist with broad competition. The private profiles contain 27 `choose_one` substitute groups, 29 `can_use_multiple` groups, and 52 complement groups.

Across the 95 accepted experimental cells, every selected good has at least two positive bidders and every cell passes the preregistered structural and economic checks. The full-information optimum has a median of three winners (mean 3.0); the largest winner contributes 59.3% of optimum welfare on average and never more than 88.4%. All selected goods are allocated in the full-information optimum. These diagnostics establish that the suite contains competition and non-additive structure rather than mechanically concentrating all value in one bidder. They describe one generated PC-component population, not a representative sample of all combinatorial-auction domains.

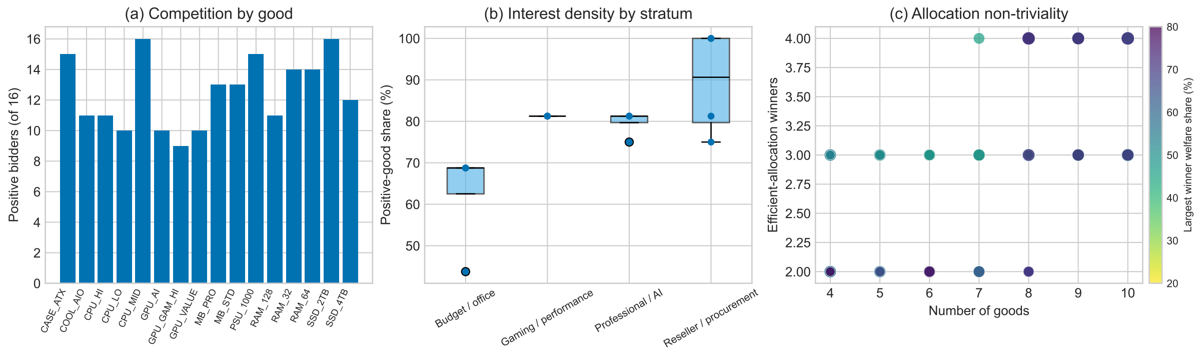

Figure 1: Structure of the generated experimental environment. Panel (a) reports positive bidders per master-population good; panel (b) reports bidder interest density by stratum; panel (c) reports full-information winner counts across accepted cells, with colour indicating the largest winner's welfare share and marker size indicating bidder count.

8.2 Provisional-Value Calibration

The out-of-domain calibration selects a uniform scale of 1.4758. Across nine held-out synthetic environments, mean normalised payment error falls from 0.289 without calibration to 0.233 with calibration, a 19.5% relative reduction; seven of nine held-out cases improve. The scale is therefore used for all initial provisional bids in the final PC-component experiments.

8.3 Interest-Map Support Reduction

Figure 2 compares the full non-empty powerset with final candidate support. Final support excludes every bundle containing an inferred-uninterested good and every bundle containing multiple members of an evidence-grounded `choose_one` group. Across all 95 cases it is 55.5% smaller than the full-support baseline. With eight bidders and ten goods, support falls from 8,184 bidder–bundle valuations to 2,619.2 candidates, a 68.0% reduction. At the 8×8 anchor, it falls from 2,040 to 757.6 on average.

This reduction changes the practical level of the communication problem but does not remove exponential worst-case growth. Its interpretation is also deliberately narrow. The opening disclosures are validated against hidden profiles before interest-map inference, so high

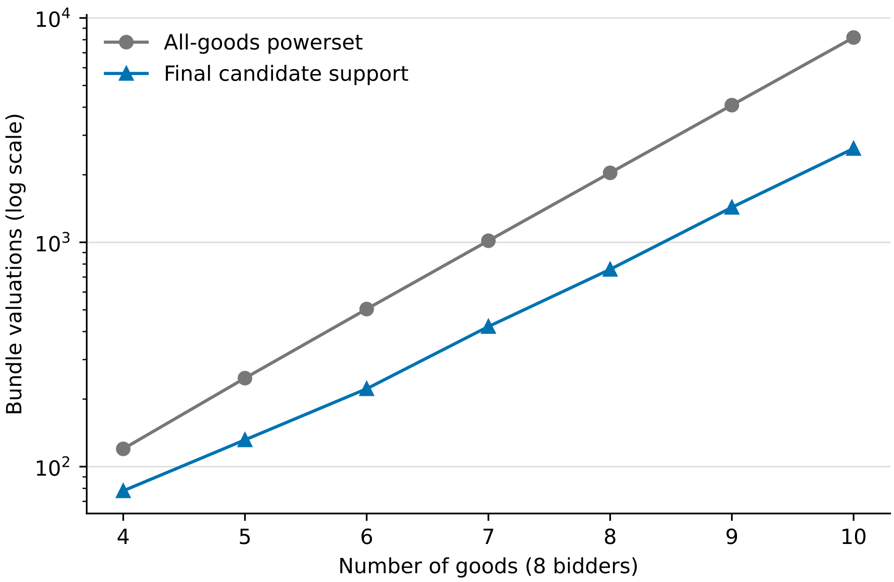

Figure 2: Auction-level candidate-support size with eight bidders as the number of goods varies. Points are five-seed means; the vertical axis is logarithmic. The full baseline is $8(2^{|G|} - 1)$.

reconstruction accuracy is a pipeline check on reconstruction from validated disclosure, not evidence of unrestricted preference recovery from arbitrary language.

8.4 Event-Policy Ablations

Policy selection uses five-seed 8×8 ablations and precedes the 95-case evaluation. In the sealed study, incumbent verification alone achieves 92.2% mean efficiency with 11.2 queries. Adding winner-removal counterfactuals raises efficiency to 95.6% with 19.0 queries; adding scarcity fallbacks reaches 97.8% with 22.4; and the full *adaptive competitive-frontier refinement* (ACFR) policy, including large-correction follow-up, reaches 99.1% with 26.2. Removing incumbent verification produces the largest degradation, to 89.4%.

The clock ablation holds the top-three, $50-step price path fixed and varies only terminal events. PV-only supplementary allocation achieves 88.1% with no exact query. A single revealed-witness treatment reaches 90.2% with three queries; winner closure reaches 92.2% with 11.2; and the selected revealed-winner sandwich reaches 95.7% with 13.4. Full frontier closure is slightly less efficient at 95.2% while using 21.4 queries. The sandwich is therefore the best observed efficiency–query trade-off among the tested policies, rather than a claim that adding events is monotonically beneficial.

A leave-one-composition-seed-out stability check provides a limited guard against selection by one favourable ordering. Maximising mean efficiency on four seeds selects ACFR in all five sealed folds, with a tie against the policy without scarcity in one fold; ACFR's held-out mean is 99.1%. The revealed-winner sandwich is selected in all five clock folds and retains 95.7% held-out mean efficiency. This supports stability across the available compositions, but it is not validation on a newly generated population.

8.5 Full-Suite Overview

Table 3 gives the main paired results. The calibrated initial bids achieve 86.4% mean efficiency. Sealed elicitation raises this to 94.8%, a paired gain of 8.4 percentage points (seed-clustered 95% CI 3.9–12.9), and improves 78 of 95 cases. Clock elicitation raises mean efficiency to 92.8%, a 6.5-point gain (95% CI 1.3–11.6), and improves 68 cases. Selective exact correction is

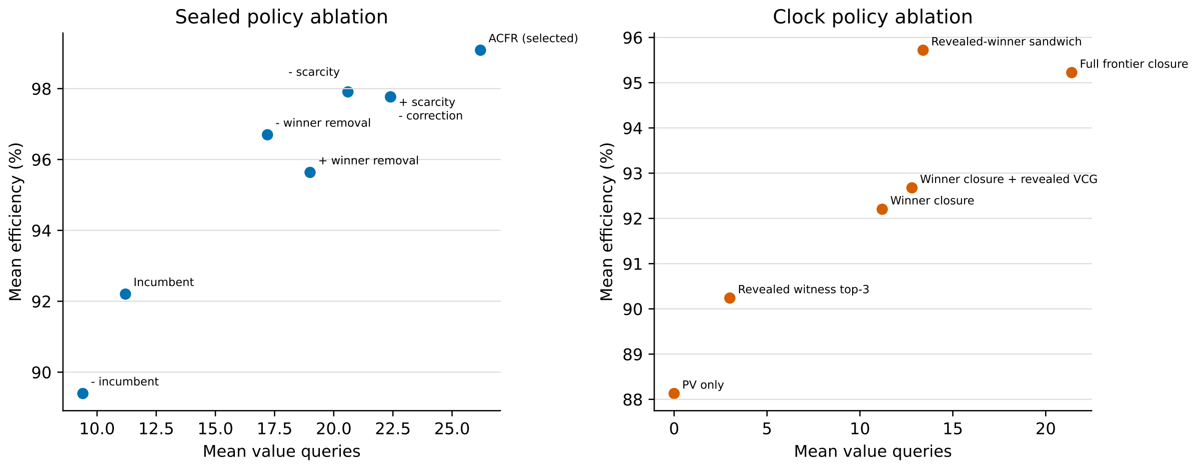

Figure 3: Five-seed 8×8 event-policy ablations. The selected policies are adaptive competitive-frontier refinement for sealed and the revealed-winner sandwich for clock.

non-monotonic: sealed worsens two cases and clock worsens 12 because the WDP continues to compare corrected atoms with remaining provisional atoms.

Table 3: Final allocation and communication results over 95 matched cases. Efficiency intervals use five seed-level means. Better/same/worse compares each elicited mechanism with its common initial provisional allocation.

Arm	Mean eff.	Median	95% CI	$\eta \geq 95\%$	Exact	Mean VQs
Initial PV	86.4%	88.2%	82.3–90.5%	19	7	0.0
Sealed	94.8%	97.8%	93.7–95.8%	57	23	21.8
Clock	92.8%	95.0%	88.8–96.8%	48	15	12.3

Figure 4 shows all three scalability paths. Mean sealed efficiency is 95.0% when bidders vary at eight goods, 94.5% when goods vary at eight bidders, and 96.0% along the joint path, with the common anchor included in each visual series. Clock gives 94.4%, 92.4%, and 92.5%, respectively. The curves are non-monotonic because successive cells are nested changes in composition rather than iid draws, but they show no collapse in average efficiency through the 10×10 endpoint.

The final event policies produce a clear query trade-off. Figure 5 shows 21.8 exact VQs per sealed case and 12.3 per clock case. Clock uses 9.6 fewer queries on average (seed-clustered 95% CI 7.4–11.7 fewer) and is less query-intensive in 86 of 95 paired cases. This is a consequence of the final clock redesign: price discovery itself makes no exact query, and only revealed terminal winners and VCG witnesses are eligible for verification. The sealed mechanism searches its allocation and counterfactual frontier during every elicitation round.

All 95 clock auctions clear through no excess demand. The median clock lasts 30 rounds, the mean 33.6, and the maximum 84. Seventeen cases require more than 50 rounds, confirming that the frozen 500-round value is a safeguard rather than a binding termination rule. Sealed elicitation stops after a median of five and a maximum of 29 refinement rounds, below its 40-round safeguard.

8.6 Sealed Outcomes and Event Path

The final suite records allocation incidence and reported-VCG-witness incidence for every exact query. In sealed ACFR, allocated bundles account for 742 queries and winner-removal counterfactuals for 683; 30.5% and 31.3%, respectively, appear in the final reported VCG witness set. Large-correction follow-ups account for 288 queries with a 27.1% witness hit rate. Scarcity

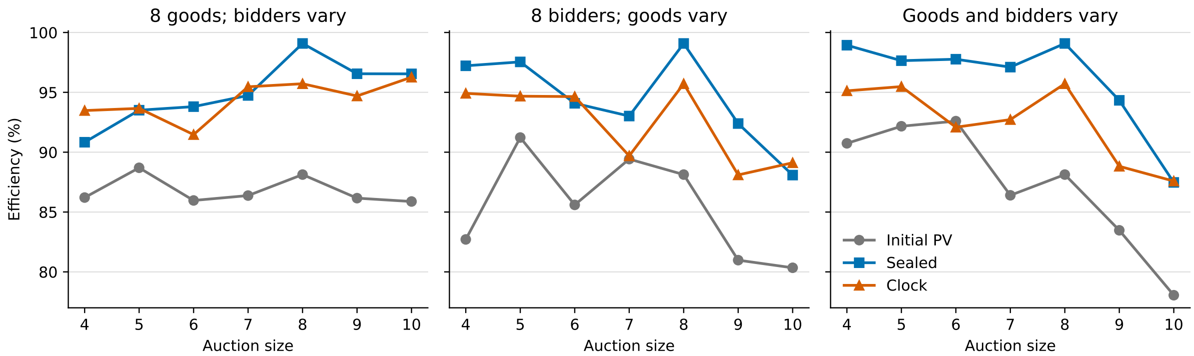

Figure 4: True allocation efficiency across the bidder, goods, and joint scalability paths. Lines show cell means over five seeds and points show seed-level observations. The 8×8 anchor appears in each panel.

Figure 5: Deterministic exact value queries across the final scalability suite. Offline disclosure, interest-map, and provisional-valuation calls are excluded.

fallbacks account for 361 queries and have a lower 11.6% direct witness rate. That lower rate does not establish zero causal value: an exact fallback can alter an intermediate allocation and expose a later winner or counterfactual.

Across all sizes, ACFR reaches 94.8% mean efficiency with 21.8 exact queries, improving 78 of 95 initial allocations. Its median of five refinement rounds reflects repeated WDP feedback rather than a fixed query schedule. In the seed-zero four-good, eight-bidder example, incumbent and counterfactual events raise efficiency from 45.7% to 96.6% in five rounds and 15 queries. This illustrates the intended sealed logic: verify the reported winner, expose its nearest allocation competitors, and re-optimise after correction.

8.7 Clock Outcomes and Event Path

In the clock mechanism, winner closure accounts for 754 queries with a 29.7% reported-witness hit rate. The staged revealed-witness phase accounts for 125 queries with a 40.8% hit rate, while the frozen single-pass revealed-witness stage is the most targeted: 57.5% of its 287 queries enter the final reported witness set. These diagnostics show how the clock's demand path restricts verification without eliminating payment-relevant bundles.

The clock reaches 92.8% mean efficiency with 12.3 exact queries and improves 68 of 95 initial allocations. Every price path clears through zero excess demand before terminal verification. Its lower query count follows directly from price-path eligibility: an unobserved losing atom cannot enter the revealed-witness frontier. The trade-off is narrower corrective coverage and 12 cases in which terminal correction lowers efficiency relative to the initial supplementary allocation.

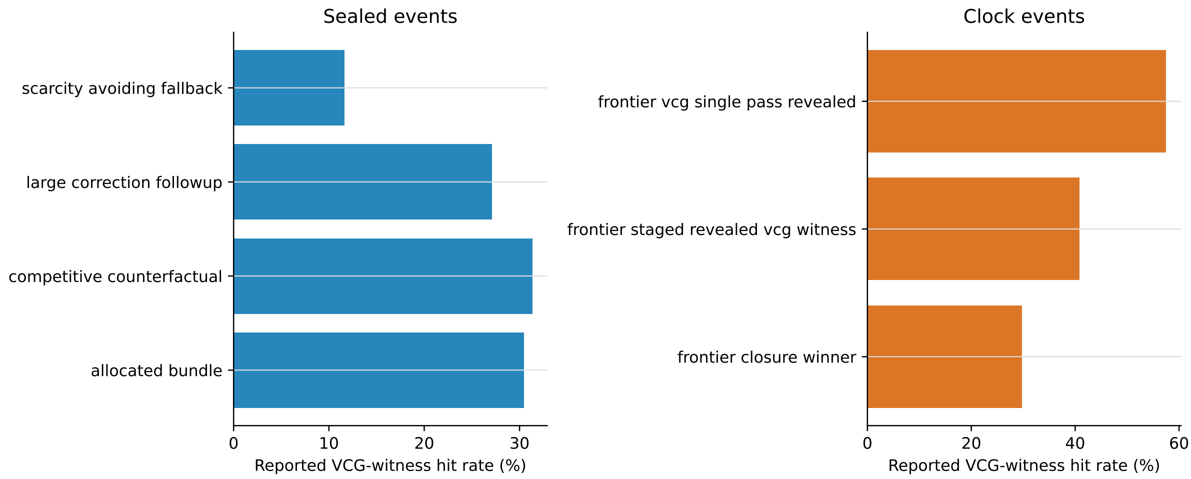

Figure 6: Reported VCG-witness incidence by event type over the final 95-case suites. Incidence is descriptive rather than a causal ablation.

Figure 7 contrasts the successful sealed path above with a clock failure. The seed-zero 6×6 clock case begins with a fully efficient supplementary allocation but terminal corrections select an allocation at 76.2% efficiency after 13 queries. Exact local corrections need not improve the global allocation while other atoms remain provisional.

8.8 Cross-Mechanism Comparison and the Payment-Counterfactual Gap

Sealed exceeds clock efficiency by 1.9 percentage points on average. Clock is higher in 21 cases, the mechanisms are equal in 22, and sealed is higher in 52. However, the seed-clustered 95% interval for clock minus sealed is –5.1 to 1.2 points, so five composition seeds do not resolve a population difference in efficiency. Figure 8 shows both the concentration near equality and several mechanism-specific failures.

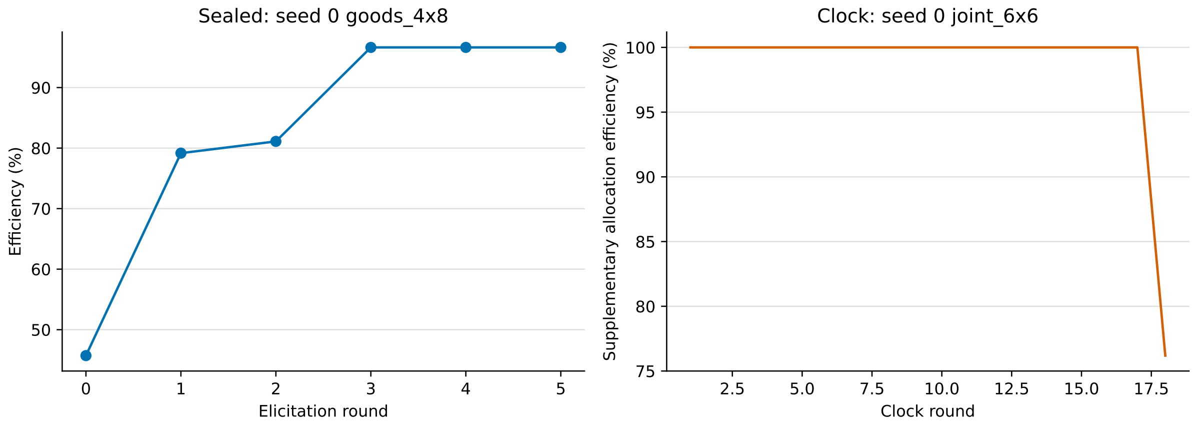

Figure 7: Illustrative final-suite trajectories: the largest sealed efficiency gain (left) and largest clock terminal welfare reduction (right). These are diagnostic examples, not representative averages.

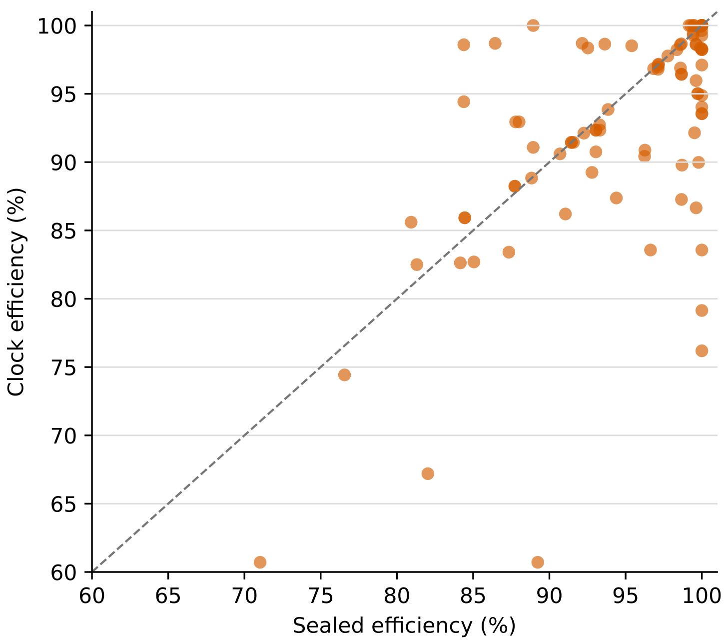

Figure 8: Paired sealed and clock efficiency under the same frozen initial proxy state. The dashed line denotes equal efficiency.

Payment reconstruction is a separate and more demanding information problem, as Table 4 makes explicit. An efficient allocation can be supported by accurate values for a small set of winning and displacing bundles. VCG additionally requires a correctly reported bidder-removal optimum for every winner.

Table 4: Different information requirements for allocation and VCG pricing.

Outcome	Required reported information	Sparse-elicitation failure
Efficient allocation	Values sufficient to rank feasible allocations near a^*	Winning or close competing bundle is omitted or misranked
Winner i's VCG payment	Allocation information plus accurate W^*_{-i}	Bidder-removal witness remains provisional, omitted, or unrevealed

Empirically, payment reconstruction remains materially weaker than allocation recovery. Mean absolute per-winner payment error, normalised by optimum welfare, is 32.1% for sealed and 36.5% for clock. The corresponding absolute revenue errors are 12.0% and 13.4%. Paired payment-error and revenue-error differences both have seed-clustered intervals crossing zero. Figure 9 shows that high allocation efficiency does not force reported-space VCG payments to approach the oracle payments: each winner's payment also depends on welfare in a bidder-removal allocation whose support may remain provisional or unrevealed.

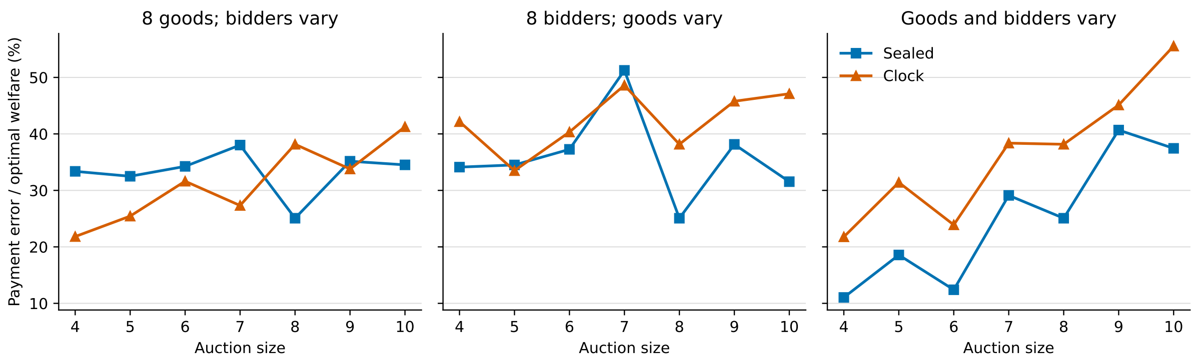

Figure 9: Absolute reported-space VCG payment error normalised by full-information optimum welfare. Lower is better.

9 Discussion

9.1 Support Selection and Provisional Scoring

The interest map and provisional valuation solve different problems. Interest-map inference is a support-selection decision: excluded items and evidence-grounded acquisition modes determine which bundles can enter the initial bid. Provisional valuation is a scoring decision conditional on that support. A candidate omitted by the interest map cannot be recovered merely by improving the PV prompt. Conversely, a broad support does not guarantee a good allocation if allocation-relevant bundles are badly ranked.

This decomposition is useful both scientifically and operationally. Exact values on interest-map support measure the best outcome available after support filtering. Comparing that benchmark with proxy PV identifies scoring loss. Comparing oracle and inferred candidate sets identifies exclusion and substitute-mode loss. It also prevents interest-map performance from being judged only through the final auction outcome, where several errors interact.

9.2 Mechanisms as Elicitation Environments

Sealed and clock auctions begin from the same frozen proxy state but expose different reasons to revisit a bundle. Competitive sealed feedback searches the reported allocation frontier directly throughout its iterative phase. The final clock policy instead uses the price path as a relevance filter: it first discovers demand without exact queries, then verifies newly exposed winners and bidder-removal witnesses only when those witnesses were revealed on the clock path. The event interface makes these differences explicit without embedding LLM orchestration inside the auction mechanism.

The ablations show that more querying is not automatically better. Full clock-frontier closure uses eight more queries than the selected sandwich yet has slightly lower mean efficiency. Direct event "hits" must nevertheless be interpreted cautiously. A queried bundle need not appear in the final or full-information allocation to be causally useful. Correcting it can change an intermediate WDP allocation, which exposes another winner or counterfactual bundle. Event evaluation therefore combines direct final-allocation incidence with trajectory and ablation evidence.

9.3 Non-Monotonic Refinement

Refinement is not guaranteed to improve true welfare monotonically. The WDP optimises a mixture of exact queried values and uncorrected provisional values. Correcting one bundle can therefore move the reported optimum to an allocation with lower hidden welfare, even though the individual correction is exact. This is a central feature of selective elicitation rather than necessarily an implementation defect. Useful query policies must be both value-correcting and allocation-relevant, and reported trajectories should not be assumed to converge monotonically.

9.4 Preparation Cost and Person-Side Query Cost

The implementation separates person-model disclosures, verification, interest-map inference, PV chunks, value queries, and demand queries. This matters because their costs have different interpretations. Person and proxy tokens are one-off offline preparation costs attached to frozen packs. Mechanism-triggered VQs are person-side preference queries, but in the principal experiment they are deterministic and consume no LLM tokens. Mechanism runtime therefore scales in WDP solves, clock rounds, and exact queries rather than hosted-model calls.

Interest-map filtering reduces the candidate powerset by 55.5% on average, but support generation still grows rapidly for broad-interest bidders. PV chunking improves structured-output reliability; it does not change the underlying number of candidate bundles that must be scored. Candidate count and preparation cost must therefore remain conceptually separate from auction-stage exact-query counts. The reported mechanism comparison concerns the latter because all offline inference is shared and amortisable.

9.5 Allocation and Payment Accuracy

Sparse elicitation places different information demands on allocation and VCG pricing. Efficient allocation requires enough information to rank feasible outcomes near the optimum. Accurate VCG payments additionally require winner-removal welfare. A large revenue difference at high allocation efficiency is therefore informative rather than contradictory: it indicates that the allocation frontier was learned more accurately than the payment counterfactual frontier. VCG-witness logging makes this distinction observable, but the residual 32–36% normalised payment error shows that the selected sparse policies do not fully close bidder-removal support.

10 Limitations

Four limitations most directly bound the paper's claims:

1. **One domain and one generated population.** The five composition seeds are nested samples from one validated 16-by-16 master population, not five independently generated populations. Statistical intervals therefore describe composition sensitivity within this population.

2. **No strategic behaviour.** Provisional estimates and exact refinements are treated as reports to be corrected, without bid shading, manipulation, or equilibrium analysis. The mechanisms are experimental elicitation environments rather than deployment-ready auction designs.

3. **No human-subject evidence.** The architecture reduces formal reporting and presents a natural-language interface, but it does not establish that real bidders experience lower cognitive burden or can answer repeated exact value queries consistently.

4. **Policy selection within the same population.** The 8×8 ablations and 95-case evaluation share the master population. Leave-one-seed-out stability helps only across compositions; it does not establish transfer to a new domain or independently generated population.

Additional modelling limitations remain. Population constraints and the deterministic valuation formula encode judgement about complements, substitutes, budgets, caps, and saturation. The fixed model assignment, validated disclosures, and closed-world interest map may overstate inference reliability relative to incomplete or strategic human language. Candidate support is frozen rather than revised online. The CCA-inspired clock omits activity rules, a full supplementary-bidding phase, optimised increments, and core-selecting payments. The current environment also assumes unit-supply indivisible goods. Model-role robustness, other mechanisms such as CECA, and independently generated domains remain future validation rather than evidence claimed here.

11 Extensions

Natural extensions include:

- dynamic interest-map updates after mechanism feedback;
- uncertainty-aware event priority based on pivotality and confidence;
- a bounded post-allocation payment-counterfactual audit;
- demand-query and noisy human-response robustness treatments;
- interval-valued or uncertainty-calibrated provisional valuations;
- CCA-lite activity rules and core-selecting payments;
- strategic proxy policies;
- independently generated populations, domains, and multi-model role robustness;
- human-subject validation of the person-simulator assumptions;
- CECA-style satisfaction and counterexample event adapters.

12 Conclusion

This paper develops a reproducible framework for scalable preference elicitation with LLM proxy bidders. A constrained generation-and-validation pipeline supplies structured private benchmark environments. A language-to-bid pipeline separates candidate selection through an interest map from candidate scoring through provisional valuations. A modular event interface then embeds the same frozen proxy state in iterative sealed and ascending clock auctions.

The separation between support, scoring, and refinement is central. It makes candidate omission distinguishable from provisional misvaluation and makes mechanism feedback comparable without repeated LLM sampling. The closed-world interest map provides scalability, while evidence-grounded acquisition modes protect multi-use bidders from overly aggressive substitute filtering. Frozen replay and deterministic value queries isolate the economic query path from hosted-model variability.

The framework does not remove the exponential communication problem for general combinatorial values. Instead, it provides a controlled empirical method for studying where information is discarded, how initial bids are formed, and which allocation- or price-derived events recover decision-relevant values. This makes the interaction between language inference and auction design observable, reproducible, and amenable to ablation.

In the final five-composition-seed evaluation, interest maps reduce candidate support by 55.5% relative to full powersets. Calibrated provisional bids achieve 86.4% mean efficiency; sealed and clock elicitation raise this to 94.8% and 92.8% with 21.8 and 12.3 exact value queries, respectively. The selected clock policy preserves genuine ascending-price discovery and clears naturally in all 95 cases, while its revealed-demand restriction substantially reduces query volume. Neither policy reconstructs VCG payments as successfully as it reconstructs allocations, identifying sparse payment-counterfactual closure as the most important remaining mechanism-design problem.

A Reproducibility and Additional Specification

A.1 Frozen Model Roles and Information Boundaries

The accepted master population was generated by OpenAI `gpt-5.6-sol`. Frozen opening answers and their independent verifier calls also use `gpt-5.6-sol`. Interest maps and provisional valuations use Google `gemini-3.6-flash`. The bidder proxy receives only the public catalogue and accepted opening exchange; it never receives latent profiles or complete valuation tables. All auction-stage exact answers are deterministic lookups and therefore use no model provider.

The five composition seeds $0, \ldots, 4$ instantiate distinct nested goods and bidder orderings from the same master population. Each goods-specific master elicitation pack is projected to smaller bidder sets, ensuring that changing the number of rivals does not regenerate a bidder's opening exchange or provisional values.

A.2 Frozen Preparation and Mechanism Parameters

Structured-output parsing is fail closed in the principal preparation. A failed interest map or provisional-value chunk is retried with diagnostic context and then aborts rather than silently expanding to all goods or replacing the treatment with mass exact queries. Successful model responses are committed independently to a read-write SQLite cache keyed by the rendered prompt and response-relevant settings. Frozen packs record raw and parsed responses, model role, token counts, latency, cache status, candidate support, and configuration hashes.

Table 5: Principal frozen parameters. Zero query caps denote disabled caps.

Component	Setting	Value
Opening interaction	Question policy	Canonical, one accepted answer
Interest map	World assumption / failure	Closed world / fail closed
Candidate support	Maximum bundles	Uncapped
Provisional valuation	Chunk / failure	64 bundles / fail closed
Parsing	Additional retries	2
Calibration	Family / scale / size factor	Uniform / 1.4758 / 1.0
Sealed	Events	Incumbent, winner removal, scarcity, correction follow-up
Sealed	Stop / safeguard	No new refinements / 40 rounds
Sealed	Correction / scarcity threshold	25% / 60% of incumbent report
Clock	Demand / price increment	Top three / $50
Clock	Stop / safeguard	No excess demand / 500 rounds
Clock	Terminal policy	Single-pass revealed witness and staged winner/witness closure
Queries	Per-bidder / global cap	0 / 0
Queries	Exact response mode	Deterministic hidden-value lookup

A.3 Frozen Artifacts and Offline Reconstruction

The experiment is anchored by `outputs/final/final_experiment_v3.json`, which records scenario and pack hashes, calibration provenance, event policy, mechanism parameters, and all 95 pack paths. Raw sealed and clock outputs are stored under `outputs/final/scalability_v2` and `outputs/final/clock_sandwich_scalability_v3`. The final analysis is reconstructed without LLM calls by:

```
./venv/bin/python scripts/build_final_analytic_package.py \
  --spec outputs/final/final_experiment_v3.json \
  --sealed-dir outputs/final/scalability_v2 \
  --clock-dir outputs/final/clock_sandwich_scalability_v3 \
  --interest-map-dir outputs/interest_map_analysis \
  --sealed-ablation outputs/final/event_policy_8x8 \
  --clock-ablation \
    outputs/final/clock_focused_closure_ablation_pc_8x8_calibrated \
  --output-dir outputs/final/analytic_package_v3
```

Generated-environment tables and Figure 1 are reconstructed from the accepted population and its validation report:

```
./venv/bin/python scripts/analyze_generated_environment.py \
  --scenario-spec \
    scenarios/pc_build_v3/pc_build_population_16x16.json \
  --validation-report \
    scenarios/pc_build_v3/pc_build_population_16x16.validation.json \
  --output-dir outputs/final/analytic_package_v3
```

The analytic package contains case-level inputs, seed-clustered summaries, event and VCG-witness diagnostics, calibration evidence, selected trajectories, environment tables, and both

PDF and PNG figures. Rebuilding these outputs does not mutate the frozen scenario, elicitation packs, or auction runs.

References

Gianluca Brero, Benjamin Lubin, and Sven Seuken. Machine learning-powered iterative combinatorial auctions, 2019.

Paul W. Goldberg, Edwin Lock, and Francisco Marmolejo-Cossío. Learning strong substitutes demand via queries. *ACM Transactions on Economics and Computation*, 2022. arXiv:2005.01496.

David Huang, Francisco Marmolejo-Cossío, Edwin Lock, and David C. Parkes. Accelerated preference elicitation with llm-based proxies, 2025.

Sébastien M. Lahaie and David C. Parkes. Applying learning algorithms to preference elicitation. In *Proceedings of the 5th ACM Conference on Electronic Commerce*, pages 180–188, 2004.

Benjamin Lubin, Sven Seuken, Manuel Beyeler, and Gianluca Brero. imlca: Machine learning-powered iterative combinatorial auctions with interval bidding, 2020.

Noam Nisan and Ilya Segal. The communication requirements of efficient allocations and supporting prices. *Journal of Economic Theory*, 129(1):192–224, 2006.

David C. Parkes. Iterative combinatorial auctions. In Peter Cramton, Yoav Shoham, and Richard Steinberg, editors, *Combinatorial Auctions*. MIT Press, 2006.

David C. Parkes and Lyle H. Ungar. ibundle: An efficient ascending price bundle auction. In *Proceedings of the 1st ACM Conference on Electronic Commerce*, 2000.

Ermis Soumalias, Jakob Weissteiner, Jakob Heiss, and Sven Seuken. Machine learning-powered combinatorial clock auction, 2023.

Ermis Soumalias, Jakob Heiss, and Jakob Weissteiner. Prices, bids, values: One ml-powered combinatorial auction to rule them all, 2024.

Ermis Soumalias, Yanchen Jiang, Kehang Zhu, Michael Curry, Sven Seuken, and David C. Parkes. Llm-powered preference elicitation in combinatorial assignment, 2025.